Assignment#1:

Preliminary Literature Review and Problem Definition

Course: Research Methods

Level: MS (CS / AI/ DS)

Assignment Title: Exploring the Literature and Defining a Research Problem

Total Marks: 20

Submission Format: Word or PDF document

Submission: 4 Weeks (2 weeks for Part-1, 2 weeks for Part-2)

1. Objectives

- Learn to search, review, and summarize relevant research literature.

- Identify gaps, limitations, or open issues from existing studies.

- Formulate a clear, evidence-based research problem based on the literature.

- Derive research questions and objectives aligned with the identified gap.

2. Task Description

This assignment is divided into two stages to ensure that the problem definition emerges logically from the literature review.

Part 1 — Preliminary Literature Exploration (10 marks)

1. Select a broad area of interest in computer science, software engineering, or artificial intelligence (e.g., software quality, AI in education, software testing, model-driven engineering, smart healthcare systems, etc.).

2. Collect and read at least 5–8 recent (last 5 years) relevant research papers from credible databases (IEEE, Springer, Elsevier, ACM, etc.).
3. For each paper, summarize briefly: purpose, methods, key findings/contributions, and limitations/gaps.
4. Synthesize your findings to describe: the current state of research in your area, common challenges or trends observed, and the specific gap that remains unexplored.

Part 2 — Defining the Research Problem (10 marks)

5. Based on the gaps found in Stage 1, formulate a precise research problem.
6. Write at least two research questions that emerge from this problem.
7. Define 3–4 research objectives that your future study could aim to achieve.
8. Clearly explain why the problem is significant and how it contributes to knowledge or practice.

3. Deliverables

- Title of the Study

- Brief Background / Context

- Summary of Literature Review (Table or Paragraph Form)

- Identified Research Gap

- Problem Definition

- Research Questions and Objectives

- References (APA or IEEE style)

4. Evaluation Criteria

Component	Marks	Description
Literature Coverage & Relevance	5	Use of current and credible studies related to topic
Identification of Research Gap	5	Logical synthesis of reviewed studies showing gap
Clarity of Research Problem	5	Problem clearly stated,

		justified, and linked to literature
Research Questions & Objectives	3	Logical and aligned with problem
Organization & Presentation	2	Proper formatting, citation, and coherence

5. Suggested Area for Exploration

- Software Quality and Microservices Architecture

- AI-Assisted Software Testing

- Model-Driven Engineering in Industry

- AI in Education

- AI Tools for Requirement Engineering

- Software Reusability in Cloud-Based Systems

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5. Suggested Topic for Exploration

1. An Exploratory Study on the Reusability Aspects of Microservices Architecture (MSA)
2. An Exploratory Study on the Quality Factors Influencing the Adoption of Microservices Architecture (MSA)
3. An Exploratory Study on the Usage of AI Tools for Software Engineering Tasks
4. An Exploratory Study on the Adoption and Usage of Model-Driven Engineering (MDE)
5. An Exploratory Study on the Detection of Design Smells in AI-Generated Software Designs
6. An Exploratory Study on the Detection of Code Smells in AI-Generated Software Codes
7. An Exploratory Study on the Effectiveness of AI-Powered Testing Tools in Improving Software Quality
8. An Exploratory Study on the Role of Generative AI in Software Design and Architecture Decision-Making

9. An Exploratory Study on the Impact of AI-Based Requirements Elicitation Tools on Software Project Success
10. An Exploratory Study on the Use of AI Models for Effort Estimation in Agile Software Development
11. An Exploratory Study on Architectural Patterns Used in Large-Scale Cloud-Native Applications
12. An Exploratory Study on the Impact of Microservices on System Maintainability and Modularity
13. An Exploratory Study on Quality Assurance Practices in Microservices-Based Systems
14. An Exploratory Study on Software Quality Trade-offs in AI-Integrated Systems
15. An Exploratory Study on the Role of DevOps Practices in Achieving Continuous Quality Improvement
16. An Exploratory Study on Challenges in Adopting Model-Driven Engineering in Developing Countries
17. An Exploratory Study on the Effectiveness of Model-Driven Approaches for System Documentation
18. An Exploratory Study on the Use of Data-Driven Decision-Making in Software Project Management
19. An Exploratory Study on Software Engineers' Perceptions of Automated Code Generation Tools
20. An Exploratory Study on Barriers to AI Adoption in the Pakistani Software Industry
21. An Exploratory Study on Organizational Readiness for AI-Driven Software Engineering Transformation
22. An Exploratory Study on Component Reuse Practices in Modern Software Development Framework
23. An Exploratory Study on Maintainability Challenges in AI-Augmented Software Projects
24. An Exploratory Study on the Impact of Artificial Intelligence Tools on Teaching and Learning Practices in Higher Education
25. An Exploratory Study on Teachers' Perceptions and Readiness toward AI Integration in Education
26. An Exploratory Study on the Use of AI-Based Learning Platforms for Personalized Education
27. An Exploratory Study on the Ethical and Pedagogical Implications of AI Use in Educational Settings
28. An Exploratory Study on the Role of Generative AI (e.g., ChatGPT) in Academic Writing and Assessment Practice
29. An Exploratory Study on the Adoption of AI-Driven Assessment Tools in Pakistani Educational Institutions
30. An Exploratory Study on the Challenges and Opportunities of AI Integration in Online and Hybrid Education